Daniel Torren Peraire

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PhD graduate in climate economics, specialised in using agent-based modelling (ABM) to study how climate policies can induce changes in consumption lifestyles and transport for a low-carbon economy

Education

PhD Environmental Science and Technology - UAB-ICTA *2021 - 2025*

- Marie Skłodowska-Curie ITN - "Modelling the social, cultural and technological evolution of low-carbon lifestyles".
- Supervised by Prof. Dr. Jeroen C.J.M van den Bergh, Prof. Dr. Ivan Savin and Prof. Dr. Enrica De Cian.
- European Joint Doctorate with PhD in Economics from Ca' Foscari University of Venice, included one year research stay.

MSci Physics - Imperial College London - 2:1 *2016 - 2020*

- Relevant modules: MSci Project, Advanced Humanities Project, Plasma Physics and Computational Physics.

Publications

An agent-based model of cultural change for a low-carbon transition

- In collaboration with Prof. Jeroen van den Bergh and Prof. Ivan Savin, studied how behavioural decarbonisation interacts with longer-term cultural change, composed of individuals with multiple behaviours that evolve due to imperfect social learning in a social network.
- Sobol sensitivity analysis performed to explore model robustness.
- Helping to generate awareness of what roadblocks may appear to deep decarbonisation.
- Published in the **Journal of Artificial Societies and Social Simulations**
- Awarded JASSS Best PhD Student Article 2025.
- Presented at: EAERE 2024, NetSci23, IC^2S^2 2023, GENED workshop 2022.

(Mis)information diffusion and the financial market

- Collaborated with Dr. Tommaso Di Francesco in building an ABM where individuals spread and exchange information about an observable stochastic component of a dividend process inside different network structures.
- We Provide a framework to account for delayed absorption of shocks that are not immediately priced as predicted by classical financial models.
- Performed model calibration on Tesla stock using Bayesian inference.
- Published in the **Journal of Economic Behavior and Organization**.

The cultural multiplier of climate policy

- Develop an ABM of innovating firms and individuals with multiple preferences for pairs of low- and high-carbon consumption goods across multiple sectors, with varying income elasticities, that evolve due to imperfect learning in a social network.
- Study effectiveness of a carbon tax to induce behavioural spillovers.
- Published in the **Journal of Environmental and Resource Economics** .
- Presented at: ESEE 2024, EPOC International Conference 2024, EAEPE 2024.

Working papers

Driving in the wrong direction? A co-evolutionary model of electric vehicle adoption

- Developed an agent-based model analyzing EV adoption with firm innovation, a used vehicle market and social imitation dynamics.
- Calibrated model on California (2001–2023) to assess the impact of policies like carbon pricing and adoption subsidies on decarbonization and policy feasibility.
- **Working paper available on SSRN**.

Summer Schools

- Amsterdam Complexity School on Climate Change - *May 2024*
- Santa Fe Institute - Complex systems - *June 2023*
- Complexity Science Hub Vienna - *January 2023*
- Universitat Autonòma de Barcelona - Game theory - *July 2022*
- Universitat de Barcelona - Behavioural and experimental economics - *June 2022*

Work Experience

Graduate Software Engineer - BJSS *March 2021 - June 2021*

- Test-driven development using pytest and great expectations libraries.
- Introduction to Agile development using scrum.
- Learnt fundamentals of using cloud services with Microsoft Azure, passing the AZ-900 exam.
- Studied data engineering and worked towards completing the Microsoft DP-203 certificate.

Imperial Visualisations *July 2018 - August 2020*

- Produced visualisations for university staff and students to demonstrate key principles in areas such as electromagnetic waves and plasma physics (see *impvis.co.uk*).
- Lead sub-group aimed at optimising the suite of visualisations created.
- Self-taught JavaScript with a strong foundation in Vue.js, HTML and CSS.

Other Research Projects

MSci Project: Computational simulation of dusty plasma crystals *2020*

- Model of dusty plasma in experimental container producing mono-layer and 3D multi-layer crystals.
- Simulation in C++ using OOP and parallelisation in OpenMP, data analysis in Python.

Advanced Humanities Project *2020*

- Title: How did the development of agricultural capitalism create the ideology of "improvement"?
- Analysis of quantitative data from primary sources regarding economic changes in the 16th and 17th centuries.

Skills and Languages

- Python, JavaScript, C++, LaTeX, Microsoft Azure, Bash and Git.
- English (Fluent), Spanish (Fluent) and Catalan (Fluent).

Interests

- Bikepacking (GBDuro - London to Edinburgh), thru-hiking (walked across New Zealand - Te Araroa, Camino de Santiago), bouldering and film photography.